

# Fatigue in chronic inflammatory demyelinating polyneuropathy

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## Abstract

**Introduction:** In this study we aimed to better understand fatigue in chronic inflammatory demyelinating polyneuropathy (CIDP) as it relates to disease activity status.

**Methods:** Patients with probable or definite CIDP were stratified into active CIDP or CIDP in remission. Assessments of fatigue, physical impairment, disability, sleepiness, sleep quality, and depression were collected.

**Results:** Of the 85 patients included in the study, 46 (54%) had active disease, whereas 39 (46%) were in remission. Fatigue was substantial in both groups, but was more severe in the active group. Use of sedating medications was a major contributor to fatigue. Sleep quality was poor in both groups, whereas depression more commonly affected those with active CIDP. Inflammatory Neuropathy Cause and Treatment disability, poor sleep quality, and higher level of depression had the greatest effect on fatigue severity.

**Discussion:** Fatigue is common in CIDP regardless of the disease activity state. Minimizing sedating medications, improving sleep quality, and managing depression may improve CIDP-associated fatigue.

## KEYWORDS

autoimmune diseases, chronic inflammatory demyelinating polyneuropathy, fatigue, quality of life

## 1 | INTRODUCTION

Chronic inflammatory demyelinating polyneuropathy (CIDP) and Guillain-Barré syndrome (GBS) are immune-mediated peripheral nerve disorders characterized by progressive or relapsing weakness or numbness. Although the motor and sensory manifestations attract the most clinical focus, 80% of patients with GBS or CIDP report fatigue as one of the three most disabling symptoms.<sup>1</sup> In GBS, fatigue can persist for years after the acute illness and does not correlate with general strength, sensory deficits, GBS disability score, or symptom duration.<sup>1-3</sup> These observations led to the concept of “post-GBS fatigue syndrome,”<sup>2</sup> implying that, like numbness and weakness, fatigue may be a residual deficit of

the acute but no longer active illness. Although similar observations have been observed in small groups of patients with CIDP, characterization of fatigue in CIDP relative to disease activity status is less well understood.

Approximately one third of CIDP patients are able to achieve immunotherapy-free remission or cure at some stage of their disease.<sup>4-6</sup> The extent to which fatigue drives treatment decisions is unknown, but may be influential if fatigue is attributed to active CIDP when it may instead be a residual finding analogous to the “post-GBS fatigue syndrome.” The aim of our study was to better define fatigue in patients with active CIDP and CIDP in remission, and to add insight into factors that contribute to fatigue in this population.

## 2 | METHODS

### 2.1 | Identification of patients

In this cross-sectional multicenter study, consecutive patients with CIDP were enrolled from Duke University, Northwestern Medical

**Abbreviations:** BDI, Beck Depression Index; BMI, body mass index; CBT-I, cognitive-behavioral therapy for insomnia; CDAS, CIDP disease activity status; CIDP, chronic inflammatory demyelinating polyneuropathy; EFNS/PNS, European Federation of Neurological Societies/Peripheral Nerve Society; ESS, Epworth Sleepiness Scale; FSS, Fatigue Severity Scale; GBS, Guillain-Barré syndrome; HPA, hypothalamic-pituitary-adrenal (axis); I-RDAS, Inflammatory Rasch-built Overall Disability Scale; INCAT, Inflammatory Neuropathy Cause and Treatment; IRB, institutional review board; MRC, Medical Research Council; PSQI, Pittsburgh Sleep Quality Index.

Center, and the University of Minnesota outpatient clinics between January 2015 and December 2017. All patients satisfied probable or definite European Federation of Neurological Societies/Peripheral Nerve Society (EFNS/PNS) CIDP diagnostic criteria.<sup>7</sup>

## 2.2 | Standard protocol approval, registration, and patient consent

Written informed consent was obtained from every study participant. This study was conducted with institutional review board (IRB) approval at each participating site.

## 2.3 | Assessment of outcome measures

Patients were stratified into active or immunotherapy-free remission status as defined by CIDP disease activity status (CDAS) criteria: CDAS 1 = "Cure," off immunotherapy for at least 5 years; CDAS 2 = "Remission," off immunotherapy for less than 5 years; CDAS 3 = "Stable active disease," on immunotherapy for at least 1 year; CDAS 4 = "Improvement," on immunotherapy for at least 3 months but less than 1 year; and CDAS 5 = "Unstable active disease," immunotherapy naive or initiated up to 3 months earlier.<sup>4</sup> Residual deficits may be present at any CDAS stage, including remission or cure. CDAS classifications 3, 4, or 5 were considered to be active, whereas those classified as CDAS 1 or 2 were considered to be in immunotherapy-free remission. All subjects underwent assessments of physical impairment, disability, fatigue, depression, sleepiness, and sleep quality at a single time-point. Outcome measures in patients with active CIDP were compared with those in immunotherapy-free remission. Sedating medications, as defined by reported potential side-effect profiles, were also recorded. We included the following categories of medications: antidepressants; antispasmodic agents; pain medications, including opioids; benzodiazepine receptor agonists, including both benzodiazepine and non-benzodiazepine agents; and anticonvulsants, specifically gabapentin and pregabalin. Immunosuppressant medications were also included in our analysis, which included use of azathioprine, mycophenolate mofetil, or rituximab.

Physical impairment was assessed via Jamar grip strength dynamometer (Lafayette Instrument, Lafayette, Indiana) and Medical Research Council (MRC) sum scale. An MRC sum score of 0 to 80 was determined using conventional MRC grade scoring in the following eight muscle pairs: bilateral shoulder abduction; elbow flexion; wrist extension; finger abduction; hip flexion; knee extension; foot dorsiflexion; and great toe dorsiflexion. The Inflammatory Rasch-built Overall Disability Scale (I-RODS) and Inflammatory Neuropathy Cause and Treatment (INCAT) disability score were used to assess disability.<sup>1,6,8</sup> Raw scores from the I-RODS scale were converted to the centile score for analysis.<sup>8</sup> Pain was recorded on a visual analog scale.

Fatigue is defined as an overwhelming sense of lack of energy or feeling of exhaustion with impaired physical or cognitive function, whereas sleepiness is defined as a propensity to fall asleep from

presumed impairment of the normal arousal mechanism.<sup>9</sup> Fatigue, depression, and sleepiness were investigated independently as well as analyzed together. The Rasch-built 7-item modified Fatigue Severity Scale (FSS) was used to assess fatigue.<sup>10</sup> The FSS is scored from 0 to 21, where higher scores indicate greater severity of fatigue. FSS scores from study participants were compared with an equal number of age- and gender-matched healthy controls recruited and consented as per institutional IRB protocol. Sleepiness was quantified using the Epworth Sleepiness Scale (ESS). ESS values of 0 to 7 were considered normal sleepiness, 8 to 9 average sleepiness, 10 to 15 excessive sleepiness, and 16 to 24 severe sleepiness.<sup>11</sup> The Pittsburgh Sleep Quality Index (PSQI) was recorded in all patients to capture sleep quality, and the Beck Depression inventory (BDI) was used to quantify depression. A PSQI of greater than 5 was considered poor sleep quality.<sup>12,13</sup> On the BDI, by conventional reference, data scores of less than 14 were considered minimal depression, 14 to 19 mild depression, 20 to 28 moderate depression, and 29 to 63 severe depression.<sup>14</sup>

## 2.4 | Statistical analysis

We performed a descriptive analysis depicting the means or proportions of continuous or categorical demographic characteristics, clinical variables, and outcome metrics among the two groups (CIDP active and CIDP remission). We used Student's two-tailed *t* test comparing means in the two groups and Pearson's chi-square and Fisher's exact tests for proportions of categorical variables among the active and remission patients. We subsequently carried out bivariate linear regressions examining the associations between patient Rasch-built FSS scores, demographics, clinical characteristics, and CIDP-related outcome characteristics in the active and remission groups. All covariates from Table 1 were included in the regression analysis. We further built multivariable linear regression equations assessing the association between outcome metrics and the dependent variable of the Rasch-built FSS, controlling for demographic and clinical covariates, as well as active vs remission status. For the purposes of the analysis here, scores on ordinal scale metrics were considered as continuous variables. All analyses were conducted with STATA version 14.0 software (StataCorp, College Station, Texas). Significance was set at *P* = .05.

# 3 | RESULTS

## 3.1 | Demographic and clinical data at enrollment

A total of 85 patients were enrolled in the study. Forty-six of these patients were considered to have active CIDP (CDAS 3 = 40, CDAS 4 = 2, CDAS 5 = 4), whereas 39 were in remission (CDAS 2 = 38, CDAS 1 = 1). Of those in remission, the mean time since immunotherapy was stopped was 21.1 months (Table 1). Of the 39 patients considered to be in remission, 13 were in remission between 6 and 12 months. Baseline demographic data and patient characteristics are shown in Table 1. Age, gender, weight, body mass index (BMI), time

**TABLE 1** Demographic and clinical data for 85 patients with CIDP

| Characteristic                        | Active (n = 46) | Remission (n = 39) | P value |
|---------------------------------------|-----------------|--------------------|---------|
| Age, in years                         | 51.9 (12.2)     | 47.7 (13.1)        | .13     |
| Male                                  | 61%             | 49%                | .27     |
| Weight (kg)                           | 82.2 (17.6)     | 85.7 (18.7)        | .38     |
| BMI                                   | 26.9 (5.2)      | 28.6 (5.6)         | .14     |
| Second autoimmune disease             | 15%             | 5%                 | .12     |
| Monoclonal gammopathy                 |                 |                    |         |
| Immunoglobulin M                      | 10.8%           | 7.7%               | .45     |
| Immunoglobulin G                      | 4.3%            | 2.6%               | .99     |
| Pre-existing depression               | 26%             | 21%                | .55     |
| Sedating medications                  | 48%             | 31%                | .11     |
| Time from symptom onset, in months    | 87.2 (80.2)     | 107.4 (71.7)       | .22     |
| Time since CIDP diagnosis, in months  | 72.4 (76.0)     | 93.6 (66.4)        | .17     |
| Time in remission, in months          | NA              | 21.1 (16.4)        | NA      |
| Immunotherapy, n (%)                  |                 |                    |         |
| Steroids <sup>a</sup> any dose        | 9 (20%)         | NA                 | NA      |
| Steroids <sup>a</sup> ≥30 mg/day      | 3 (6%)          |                    |         |
| IVIg                                  | 37 (80%)        |                    |         |
| Other immunosuppressants <sup>b</sup> | 7 (15%)         |                    |         |

Note: Data expressed as mean (standard deviation), percent, or number (percent).

Abbreviations: BMI, body mass index; CIDP, chronic inflammatory demyelinating polyneuropathy; IVIg, intravenous immunoglobulin; NA, not applicable.

<sup>a</sup>Prednisone or corticosteroid at equivalent prednisone dose.

<sup>b</sup>Immunotherapies include rituximab, azathioprine, and mycophenolate mofetil.

since CIDP onset, time since CIDP diagnosis, monoclonal gammopathy, and pre-existing depression were similar in the active and remission groups. There was no significant difference between the frequency of secondary autoimmune diseases and exposure to sedating medications.

### 3.2 | Fatigue, sleepiness, sleep quality, and depression in CIDP patients

Assessments of disability and strength impairment are shown in Table 2. Patients with active CIDP had a greater degree of strength impairment as assessed by grip strength dynamometer and MRC sum score and a greater degree of disability as assessed by the I-RODS and INCAT disability scales when compared with patients with CIDP in immunotherapy-free remission. Pain severity was similar between groups. Fatigue (FSS score) was more severe in patients with active CIDP compared with those in remission, but was substantial in both groups.

In Figure 1, mean values for sleepiness, sleep quality, and depression outcomes are presented relative to reference normative data. Fatigue is presented relative to age- and gender-matched healthy controls collected as part of the current study. Healthy control patients showed a mean FSS score of 3.6 (standard deviation [SD], 3.5). FSS in both CIDP groups was more than 2 SD of the control group. There was no difference in the degree of sleepiness (ESS score) in active and

remission groups, and in both would be considered “average.” Excessive or greater sleepiness (ESS >10) was reported in 39.1% of active CIDP patients and 20.5% of inactive patients. In addition, in the active CIDP group, 39% of patients reported only fatigue, whereas 52% of patients reported an overlap of sleepiness and fatigue. Of the patients in the remission group, 46% reported only fatigue, 5% reported only experiencing sleepiness but not fatigue, and 33% reported an overlap of both sleepiness and fatigue. Active CIDP patients had worse sleep quality (PSQI score) than remission patients, and would be considered “poor” in both groups. Poor sleep quality (PSQI score >5) was present in 73.9% of active CIDP and 64.1% of inactive CIDP. Overall, 28.2% of patients reported mild or greater depression. Patients with active disease also had greater degrees of depression (BDI score) than patients in remission, with mild or greater depression (BDI score ≥14) present in 45.6% of active patients vs 7.7% of patients in remission. With respect to fatigue severity scores, compared with healthy controls, both active and remission groups had significantly more fatigue overall ( $P = .0001$ ). In addition, the active disease group had significantly more fatigue than the remission group (Table 2).

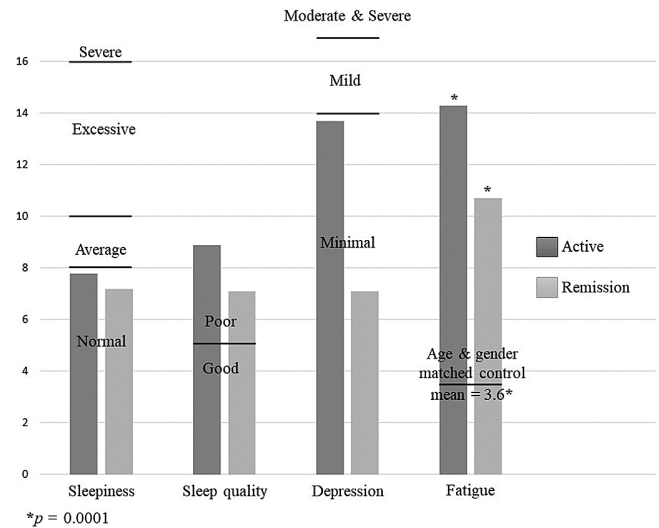
### 3.3 | 3.3 Predictors of fatigue in patients with CIDP

Bivariate linear regression analysis (Table 3) showed that, among the demographic and clinical variables, consumption of sedating

medications was associated with worse fatigue for both active and inactive CIDP. No correlation was observed for gender; weight; BMI; disease duration; or presence of pre-existing depression, monoclonal gammopathy, or another autoimmune disease. Age was associated with worse fatigue in the remission group, but not the active group.

Analysis of sleep, depression, disability, and strength outcomes (Table 3) revealed that most were associated with worsening fatigue in the active group, remission group, or both. Of the outcomes, disability, as indicated by INCAT, had the strongest correlation with fatigue. The I-RODS demonstrated a lower correlation with the FSS than the INCAT. Sleep quality and depression also correlated with INCAT but to a lesser degree. Measures of strength impairment and pain had a weaker effect on fatigue.

The multivariable linear regression analysis assessed the strength of effect of each independent outcome variable to FSS, controlling for active vs remission status and demographic and clinical covariates (Table 4). Variables that affected the FSS from strongest to weakest were higher (worse) INCAT disability, higher (worse) sleep quality, higher (worse) depression, higher (worse) sleepiness, lower (worse) I-RODS disability, lower (worse) MRC, higher (worse) pain, and lower (worse) right and left grip strength. Analysis of variance showed an



**FIGURE 1** Mean values in outcome measures in active and inactive chronic inflammatory demyelinating polyneuropathy compared with standardized normative data. BDI, Beck Depression Inventory; ESS, Epworth Sleepiness Scale; FSS, Rasch-built 7-item modified Fatigue Severity Scale; PSQI, Pittsburgh Sleep Quality Index. \*FSS in active and remission groups vs controls,  $P = .0001$

|                               | Active (n = 46) | Remission (n = 39) | P value |
|-------------------------------|-----------------|--------------------|---------|
| Grip strength (kg) left hand  | 23.1 (15.8)     | 29.8 (14.2)        | .04     |
| Grip strength (kg) right hand | 25.5 (15.6)     | 31.3 (14.8)        | .07     |
| MRC sum scale for strength    | 70.3 (11.4)     | 77.0 (4.7)         | <.01    |
| I-RODS                        | 59.1 (19.4)     | 74.9 (17.4)        | <.01    |
| INCAT disability              | 4.1 (2.04)      | 2.7 (1.5)          | <.01    |
| Pain (VAS)                    | 2.0 (2.4)       | 2.1 (2.5)          | .91     |
| <b>FSS</b>                    | 14.3 (7.0)      | 10.7 (7.0)         | .02     |
| ESS                           |                 |                    |         |
| Overall                       | 7.8 (4.4)       | 7.2 (5.0)          | .56     |
| Score 0-7                     | 50%             | 61.5%              | .38     |
| Score 8-9                     | 10.8%           | 17.9%              | .37     |
| Score 10-15                   | 32.6%           | 7.8%               | <.01    |
| Score 16-24                   | 6.5%            | 12.8%              | .46     |
| PSQI                          |                 |                    |         |
| Overall                       | 8.9 (4.3)       | 7.1 (3.9)          | .03     |
| Score >5                      | 76.1%           | 64.1%              | .24     |
| BDI                           |                 |                    |         |
| Overall                       | 13.7 (8.8)      | 7.1 (5.0)          | <.01    |
| Score <14                     | 47.8%           | 92.3%              | <.01    |
| Score 14-19                   | 26.1%           | 7.7%               | .04     |
| Score 20-28                   | 17.4%           | 0%                 | <.01    |
| Score 29-63                   | 3.3%            | 0%                 | .49     |

Note: Data expressed as mean (standard deviation) or percent.

Abbreviations: BDI, Beck Depression Inventory; BMI, body mass index; ESS, Epworth Sleepiness scale; FSS, Rasch-built 7-item modified Fatigue Severity Scale; INCAT, Inflammatory Neuropathy Cause and Treatment; I-RODS, Inflammatory Neuropathy Rasch-built Overall Disability Scale; MRC, Medical Research Council; PSQI, Pittsburgh Sleep Quality Index; VAS, visual analog scale.

**TABLE 2** Assessment of strength impairment, disability, fatigue, and sleepiness in patients with CIDP

**TABLE 3** Association between sociodemographic features and clinical outcomes with fatigue (FSS) in patients with active CIDP and patients with CIDP in remission

|                                  | Active (n = 46) |              |       | Remission (N = 39) |              |       |
|----------------------------------|-----------------|--------------|-------|--------------------|--------------|-------|
|                                  | $\beta$         | 95% CI       | P     | $\beta$            | 95% CI       | P     |
| Demographic data                 |                 |              |       |                    |              |       |
| Age                              | −0.1            | −0.3 to 0.0  | .14   | 0.02               | 0.1 to 0.4   | .001  |
| Female                           | 1.5             | −2.7 to 5.7  | .48   | 3.6                | −0.7 to 7.9  | .10   |
| Weight                           | 0               | −0.1 to 0.2  | .41   | 0                  | −0.2 to 0.1  | .49   |
| BMI                              | 0.2             | −0.2 to 0.6  | .32   | 0                  | −0.4 to 0.4  | .95   |
| Clinical features                |                 |              |       |                    |              |       |
| Second autoimmune disease        | −1.0            | −6.7 to 4.7  | .74   | 4.5                | −5.5 to 14.5 | .38   |
| Monoclonal gammopathy            | 1.4             | −4.3 to 7.1  | .63   | −4.4               | −11.7 to 2.8 | .23   |
| Pre-existing depression          | 2.5             | −2.1 to 7.1  | .29   | 4.3                | −1.1 to 9.6  | .12   |
| Sedating medications             | 3.9             | 0.0 to 7.9   | .05   | 8.1                | 4.0 to 12.2  | <.001 |
| Symptom duration (months)        | 0               | 0.0 to 0.0   | .48   | 0                  | −0.0 to 0.0  | .30   |
| Outcome assessments              |                 |              |       |                    |              |       |
| Grip strength (kg) right hand    | −0.1            | −0.2 to 0.0  | .11   | −0.2               | −0.3 to −0.0 | .02   |
| Grip strength (kg) left hand     | −0.1            | −0.2 to 0.0  | .12   | −0.2               | −0.4 to −0.1 | .004  |
| MRC sum scale score for strength | −0.2            | −0.4 to 0.0  | .03   | −0.4               | −0.8 to 0.1  | .12   |
| I-RODS                           | −0.2            | −0.3 to −0.2 | <.001 | −0.3               | −0.4 to −0.2 | <.001 |
| INCAT disability                 | 1.5             | 0.5 to 2.4   | .002  | 2.5                | 1.2 to 3.8   | <.001 |
| Pain (VAS)                       | 0.1             | 0.0 to 0.2   | .02   | 0.1                | 0.1 to 0.2   | <.001 |
| ESS                              | 0.3             | −0.1 to 0.8  | .16   | 0.6                | 0.1 to 1.0   | .009  |
| PSQI                             | 0.9             | 0.5 to 1.3   | <.001 | 0.7                | 0.2 to 1.3   | .008  |
| BDI                              | 0.5             | 0.3 to 0.7   | <.001 | 0.9                | 0.6 to 1.3   | <.001 |

Abbreviations: BDI, Beck Depression Inventory; BMI, body mass index; ESS, Epworth Sleepiness Scale; FSS, Rasch-built 7-Item modified Fatigue Severity Scale; INCAT, Inflammatory Neuropathy Cause and Treatment; I-RODS, Inflammatory Neuropathy Rasch-built Overall Disability Scale; MRC, Medical Research Council; PSQI, Pittsburgh Sleep Quality Index; VAS, visual analog scale.

**TABLE 4** Significant predictors of worse fatigue (FSS) in CIDP patients using multiple linear regression analysis

|                               | $\beta$ | 95% CI       | P     |
|-------------------------------|---------|--------------|-------|
| INCAT disability score        | 1.5     | 0.7 to 2.3   | <.001 |
| PSQI                          | 0.7     | 0.4 to 1.1   | <.001 |
| BDI                           | 0.6     | 0.4 to 0.7   | <.001 |
| ESS                           | 0.5     | 0.2 to 0.8   | .002  |
| I-RODS                        | −0.3    | −0.3 to −0.2 | <.001 |
| MRC sum scale for strength    | −0.2    | −0.4 to 1.1  | .01   |
| Pain (VAS)                    | 0.1     | 0.0 to 0.2   | .01   |
| Grip strength (kg) right hand | −0.1    | −0.3 to 0.0  | .02   |
| Grip strength (kg) left hand  | −0.1    | −0.3 to 0.0  | .02   |

Abbreviations: BDI, Beck Depression Inventory; BMI, body mass index; CIDP, chronic inflammatory demyelinating polyneuropathy; ESS, Epworth Sleepiness Scale; FSS, Rasch-built 7-item modified Fatigue Severity Scale; INCAT, Inflammatory Neuropathy Cause and Treatment; I-RODS, Inflammatory Neuropathy Rasch-Built Overall Disability Scale; MRC, Medical Research Council; PSQI, Pittsburgh Sleep Quality Index; VAS, visual analog scale.

$R^2 = 0.82$  for the multivariable combination of independent outcome variables controlling for the covariates demonstrating that a substantial proportion of the variance in the FSS was predicted by the model.

## 4 | DISCUSSION

Within this cross-sectional study of patients with well-defined CIDP we have shown that, although fatigue severity is greatest in patients with active disease, it is common even in patients in immunotherapy-free remission. As is the case for GBS in which fatigue persists as a “post-GBS fatigue syndrome,” fatigue in CIDP should be viewed as a residual symptom, not unlike numbness or weakness that may persist long after the active phase of the disease has passed.<sup>1</sup> This observation has major clinical implications, as the presence of fatigue does not necessarily mean that the disease requires or will respond to immunotherapy, but still may deleteriously affect patients as a residual persistent symptom.

Unlike monophasic GBS, in which clinical features can typically be used to differentiate the active stage from the recovery phase, the clinical boundaries that define CIDP disease activity are more ambiguous and can be difficult to identify. There is no known biomarker that can reliably predict CIDP disease activity status. The ambiguous nature of the clinical boundaries may result in a tendency to interpret CIDP motor or sensory deficits as indicative of ongoing or active disease, when they may be better attributed to residual deficits from a condition that is no longer active. Such interpretation of residual symptoms may be one reason that overtreatment of patients with CIDP is common. Randomized blinded clinical trials of CIDP have consistently shown that about 40% of patients randomized to placebo are able to reduce or stop their primary immunotherapy without deterioration.<sup>5,6</sup> It was beyond the scope of our study to determine specific reasons for immunotherapy overutilization. However, our findings do show that fatigue is common and disabling at all stages of disease. Because patients in immunotherapy-free remission can still demonstrate a significant level of fatigue when compared with healthy controls, we caution against using persistent residual fatigue as the sole justification for chronic immunotherapy utilization.<sup>1</sup>

We attempted to better understand factors that contribute to fatigue in patients with CIDP, and, by extension, rationalize therapeutic approaches that may be beneficial. Fatigue and sleepiness have overlapping features, and both can covary with poor sleep quality and depression. We found that patients with active and inactive CIDP had similar levels of sleepiness that would be considered “average,” but that sleep quality was poor in both groups. These findings give insight into potential causes of fatigue in CIDP and supportive treatment strategies. The observation that patients in both active and inactive groups did not report excessive daytime sleepiness yet they did report excessive fatigue suggests the presence of autonomic hyperarousal due to increased activation of the hypothalamic-pituitary-adrenal (HPA) axis.<sup>15,16</sup> This is akin to the fatigue and lack of daytime sleepiness in patients with chronic insomnia and what has been reported previously in chronic autoimmune diseases such as rheumatoid arthritis, myasthenia gravis, and CIDP.<sup>17-19</sup> From a treatment standpoint, these observations suggest that improved sleep hygiene and regular aerobic physical activity may be effective management strategies.<sup>20-23</sup> Exercise, in particular, has been shown to improve insomnia-related symptoms such as fatigue. It is postulated to be due to the beneficial impact of exercise on tension and anxiety and associated cognitive hyperarousal.<sup>24</sup> Studies from patients with inflammatory neuropathies (GBS and CIDP) that explored the impact of exercise on fatigue have reported mixed results. Although some have shown a beneficial impact of fatigue following a structured exercise program, others did not show an improvement in fatigue.<sup>25</sup> The discrepancy in these findings may speak to the heterogeneous causes of fatigue in inflammatory neuropathies, and hence support a multifactorial approach to fatigue management. Exposure to sedating medications was frequent in our population of patients. Although utilization of such medications is often justifiable, minimization of sedating medications when medically appropriate is encouraged for all patients, as this may be one modifiable factor that contributes to chronic fatigue in patients across

all disease activity statuses. Improved sleep hygiene with exposure to timed light therapy and cognitive-behavioral therapy for insomnia (CBT-I) also may be beneficial, as these interventions have been shown to improve both poor sleep quality and fatigue in autoimmune conditions.<sup>26-29</sup>

Depression has not been well characterized in CIDP but has been reported to occur in 9% of Dutch patients<sup>30</sup>; 12% of German patients<sup>31</sup>; and 31% of patients assessed in Serbia, Bosnia and Herzegovina, and Montenegro.<sup>32</sup> Our findings provide insight into the frequency of depression in patients from the United States, and also further characterize depression as a function of active disease. Although overall 28% of our patients were classified as mild depression or greater, in the active group 45% would be considered depressed compared with only 8% in the remission group. These findings are in line with what is known about depression in chronic medical diseases. The occurrence of depression in primary care patients ranges between 5% and 10%,<sup>33</sup> whereas in chronic illness rates tend to be two- to threefold higher. Depression in diabetes, for example, has been estimated to be 12% to 18%,<sup>34</sup> whereas depression in coronary heart disease is between 15% and 23%.<sup>35</sup> It has been demonstrated that, when depression occurs with active medical disease, it is potentially treatable even if the underlying chronic medical condition is not treatable. Just as depression contributes to the burden of disease in CIDP, so too does identification and management have the potential to improve functionality, quality of life, and adaptation to the chronic medical illness, even in the setting of chronic residual neurologic deficits.<sup>36</sup>

Our study has several limitations. Although we attempted to better understand the impact of fatigue in stages of CIDP, the co-occurrence of fatigue, sleepiness, and depression in practice is common. From a causation standpoint the relative contributions that any of these factors have on fatigue can be challenging to disentangle. Our specific outcomes with well-established normative data and a statistical approach were selected to clarify the underlying contributing factors to fatigue. We recognize the limitations acquired from utilization of the Rasch-built FSS score rather than the more common 9-item Fatigue Severity Scale.<sup>1</sup> We elected to use the Rasch-built score because of its superior clinometric properties, although we acknowledge that the results of the Rasch-built FSS are not transferable to scoring on the 9-item FSS scale. To mitigate this limitation and to accommodate for the absence of Rasch-built FSS historical control data, we collected a group of age- and gender-matched controls and herein presented the results of normative Rasch-built FSS data. We also recognize that our study is methodologically limited by the statistical analysis, which required some ordinal metrics (eg, INCAT) to be considered as continuous variables. One curious observation was that I-RODS demonstrated a lower correlation with FSS than INCAT. The reasons for this observation require additional study, but may reflect the inherently different qualities of the scales, and implies that both metrics are necessary to utilize when assessing patients with CIDP during clinical practice. Our study is further limited by a potential selection bias given the consecutive manner of recruitment of patients for the study, as patients with inactive disease may not



follow-up in clinic as often as those with active disease. Finally, we did not take into consideration how non-autoimmune chronic medical conditions and associated medications (including corticosteroids in the active CIDP group) could impact fatigue and sleepiness. It has been reported that patients who are on high-dose glucocorticoid treatment, particularly for short periods of less than 2 weeks, had a higher incidence of insomnia; however, a small percentage of our patients were on long-term glucocorticoid treatment and an even smaller percentage were on high-dose therapy. Thus, it is less likely that glucocorticoid therapy would have a significant impact (Table 1).<sup>37,38</sup>

Our findings show that fatigue is common in patients with both active and inactive CIDP. It is especially important to recognize that fatigue may persist as a residual symptom even when the disease is in remission. We encourage heightened clinical vigilance of this residual fatigue, as its presence should direct the clinician to supportive management strategies. Interventions that may be particularly helpful include: (a) minimization of sedating medications; (b) safe and ability-appropriate aerobic physical activity; (c) improvement of sleep quality via morning light therapy and cognitive-behavioral therapy; and (d) depression management, especially in the active disease state.<sup>29,39,40</sup> We anticipate that improved recognition and management of fatigue in patients with CIDP will lessen the overutilization of immunotherapy while concurrently improving CIDP-related disability.

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## CONFLICT OF INTEREST

K.L.G. is an advisory board consultant for Stealth BioTherapeutics and Akcea. J.A.A. is an advisory board consultant for Alexion and Argenx SE; has received a speaker's honorarium from CSL Behring; is a consultant for CSL Behring, Akcea, Argenx, Kaba Infusions, and Biotest; and is on the speakers' bureau for CSL Behring and Akcea. H.A. is on the editorial board for Gender and Genome, and has received royalties from Springer, a speaker's honorarium from Pre-Med, commercial research support from Fusion, and government research support from NINDS (site principal investigator on U01NS099043).

## ETHICAL PUBLICATION STATEMENT

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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